Posterior Variance Collapse in Conditional Flow Matching
is the Zero-Bandwidth Limit of a Kernel-Weighted Empirical Conditional Flow

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Abstract

Conditional flow matching (CFM) trained to convergence on a finite dataset is often described as "memorising" the training data, and overtraining is reported to collapse the conditional generative law onto individual training examples. We give an exact population-level account of this phenomenon. For a fixed dataset and the deterministic linear interpolant, the L^2-optimal velocity field is a posterior-weighted average of the per-example collapsed fields; under hard conditioning with identifying labels it reduces to a single atom, so the exact minimiser has *zero* conditional variance and zero loss (Proposition 2). Label smoothing—adding Gaussian noise to the conditioning variable—turns this hard selection into *exact kernel regression over the training atoms*: the generated endpoint law is a Nadaraya–Watson mixture $\sum_i p_i^{(h)}(y)\,\delta_{x^i}$ (Theorem 1), whose moments are a parameter-free reference curve computable from the training set (Proposition 5). We prove that this "remedy" can only match moments, never recover the posterior: the generated law stays atomic for every bandwidth h, giving an h-independent Wasserstein floor (Proposition 6). We further separate two superficially similar remedies: label noise changes the population endpoint, whereas interpolant noise provably does not, for any noise level (Proposition 8). Numerically, across five seeds, a trained MLP tracks the kernel reference to within seed noise (ratio-to-kernel ≈ 1.00 for $h \geq 0.05$), and the same collapse is reproduced on a Gaussian-mixture posterior and on MNIST inpainting. We are careful throughout to state which observations are exact population consequences and which are empirical findings the population theory does not determine.

1 Introduction

Flow matching and stochastic-interpolant models are usually trained by regressing a velocity field against a linear path between a noise sample and a data sample. On a finite dataset, exact minimisation of this objective against the *empirical* data law is known to induce memorisation, and a recent line of work has documented that conditional flow-matching models, trained past the point of good posterior calibration, produce increasingly deterministic (low-variance) conditional samples that coincide with individual training examples [phy, 2026, gra, 2025].

This paper explains the mechanism exactly. Fixing the training set $\mathcal{D} = \{(x^i, y^i)\}_{i=1}^N$ and drawing only the interpolation randomness, we characterise the exact population minimiser of the flow-matching objective over all square-integrable vector fields and the endpoint law of its flow. The

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observation that the empirical minimiser memorises is *not* new; it is established in the diffusion-memorisation literature [Pidstrigach, 2022, Gu et al., 2023, Kadkhodaie et al., 2024, Biroli and Mézard, 2023, Kamb and Ganguli, 2024]. Our contribution is the *conditional and kernel-regression structure*:

1. **Conditional sharpening.** Hard conditioning turns full-empirical-measure memorisation into *single-atom* memorisation—zero conditional variance (Corollary 1). The correct contrast with the unconditional model is not "memorises vs. does not" but *which* empirical measure is memorised.

2. **Label smoothing ≡ kernel regression.** Adding label noise makes the population endpoint an exact Nadaraya–Watson mixture over training atoms (Theorem 1), converting a 0/1 statement into a quantitative, parameter-free reference curve $\mathrm{Cov}_h(y)$ (Proposition 5).

3. **Separation of two remedies.** Label noise changes the population endpoint; interpolant noise provably does not, for any noise level (Propositions 8 and 9).

4. **Atomicity obstruction.** The smoothed law is atomic for every h, yielding an h-independent Wasserstein floor (Proposition 6): matching a second moment is not recovering the posterior.

5. **Representation floor.** A uniform Lipschitz bound on the velocity field gives a measurable loss floor $d/(3L)$ (Corollary 3) that separates representation error from optimisation error.

In one line: *conditional collapse is the zero-bandwidth limit of a kernel-weighted empirical conditional flow.*

A word on scope, developed formally in Section 3.5. Everything in the theory characterises the exact population minimiser for the empirical data law; it is not a statement about what a finite network trained by SGD attains. We therefore distinguish sharply between quantities the population theory determines (and which our experiments confirm) and quantities it does not (dependence of observed collapse on dataset size or observation noise), which we report as empirical findings rather than "predictions."

2 Setup and notation

Fix a training set $\mathcal{D} = \{(x^i, y^i)\}_{i=1}^N$ with $x^i \in \mathbb{R}^d$, $y^i \in \mathbb{R}^k$, empirical joint law $\widehat{\rho}_{XY} = \frac{1}{N}\sum_i \delta_{(x^i,y^i)}$ and marginal $\widehat{\rho}_X = \frac{1}{N}\sum_i \delta_{x^i}$. The training set is fixed (deterministic); randomness comes only from

$$I \sim \mathrm{Unif}\{1, \dots, N\}, \qquad X_0 \sim \pi_0 = \mathcal{N}(0, I_d), \qquad t \sim \mathrm{Unif}(0,1),$$

drawn independently, together with label noise $\varepsilon \sim \mathcal{N}(0, I_k)$ when $h > 0$. Set $X_1 = x^I$ and, for the deterministic linear interpolant,

$$X_t = (1-t)X_0 + tX_1, \qquad U := \dot{X}_t = X_1 - X_0.$$

The smoothed label is $\widetilde{Y} = y^I + h\varepsilon$ with Gaussian kernel $K_h(u) = (2\pi h^2)^{-k/2}\exp(-|u|^2/2h^2)$. The objective is

$$L_h(v) = \mathbb{E}\big[\,|U - v(X_t, t, \widetilde{Y})|^2\,\big]. \tag{1}$$

For $h = 0$ we write $\widetilde{Y} = Y = y^I$ and L_0. Let $\bar{x} = \frac{1}{N}\sum_i x^i$ and $\widehat{\Sigma}_X = \frac{1}{N}\sum_i (x^i - \bar{x})(x^i - \bar{x})^\top$. In the linear-Gaussian instance used experimentally, $y = Ax + \eta$ with $\eta \sim \mathcal{N}(0, \sigma_{\mathrm{obs}}^2 I_k)$, which yields a Gaussian analytic posterior with covariance Σ_{post}.

Scope marker. Everything in Sections 3.1–3.3 characterises the exact population minimiser over L^2-measurable vector fields for the empirical data law. Section 3.4 quantifies one reason a finite trained network differs.

3 Theory

We state the load-bearing results with proofs; auxiliary lemmas are proved in Appendix A. The projection identity underlying everything is standard.

Proposition 1 (L^2 projection)**.** *For every square-integrable v, $L_h(v) = \mathbb{E}[\mathrm{Var}(U \mid X_t, t, \widetilde{Y})] + \mathbb{E}\,|v - \mathbb{E}[U \mid X_t, t, \widetilde{Y}]|^2$, so the a.e.-unique minimiser is $v_h^\star(x,t,y) = \mathbb{E}[U \mid X_t = x, t, \widetilde{Y} = y]$ with $\inf_v L_h = \mathbb{E}[\mathrm{Var}(U \mid X_t, t, \widetilde{Y})]$.*

Two facts make the conditional expectations legitimate and the flows well posed without any additional assumption (Appendix A): (i) for $t < 1$ the map $X_0 \mapsto X_t = (1-t)X_0 + tx^i$ is an invertible affine change of variables, with $X_0 = (X_t - tx^i)/(1-t)$ and conditional density $p(x \mid I = i, t) = (1-t)^{-d}\pi_0\big(\frac{x-tx^i}{1-t}\big) > 0$ for every x (Lemma 1); (ii) any field of the form $v(x,t) = \sum_i w_i(x,t)\,(x^i - x)/(1-t)$ with smooth weights summing to one is C^∞ hence locally Lipschitz on $\mathbb{R}^d \times [0,1)$, so its ODE has a unique solution up to $t = 1$ and endpoint laws are the well-defined weak limits $p_1 = \lim_{t\uparrow 1} p_t$ (Lemma 2).

3.1 Hard conditioning: exact collapse

Proposition 2 (Exact conditional collapse; $h = 0$)**.** *Assume $y^1, \ldots, y^N$ are pairwise distinct. Then (a) $v_0^\star(x,t,y^i) = (x^i - x)/(1-t)$ for $t < 1$; (b) $\inf_v L_0(v) = 0$; (c) the ODE $\dot{x}_t = (x^i - x_t)/(1-t)$ has the explicit solution $x_t = (1-t)x_0 + tx^i$, so $x_t \to x^i$ for every x_0; and (d) $p_1^{\mathrm{cond}}(\cdot \mid y^i) = \delta_{x^i}$, hence $\mathrm{Cov}[p_1^{\mathrm{cond}}(\cdot \mid y^i)] = 0$.*

Proof. (a) Distinctness makes $\{Y = y^i\}$ identify $I = i$, so $X_1 = x^i$ is deterministic and, by Lemma 1, $X_0 = (x - tx^i)/(1-t)$ is determined by (x,t,i). Hence $U = x^i - X_0 = (x^i - x)/(1-t)$ is $\sigma(X_t, t, Y)$-measurable and equals its conditional expectation. (b) $U - v_0^\star = 0$ a.s. (c) Multiplying by $(1-t)$ gives $\frac{d}{dt}\big[\frac{x_t - x^i}{1-t}\big] = 0$, so $\frac{x_t - x^i}{1-t} \equiv x_0 - x^i$. (d) The endpoint is x^i independently of x_0. □

If labels are not distinct, collapse is onto the label's empirical conditional support: for $\mathcal{I}_y = \{i : y^i = y\}$, $p_1^{\mathrm{cond}}(\cdot \mid y) = |\mathcal{I}_y|^{-1}\sum_{i \in \mathcal{I}_y}\delta_{x^i}$. Collapse is always to the empirical conditional law, which is a single atom exactly when the label is an identifier.

Proposition 3 (Unconditional endpoint is the full empirical law)**.** *For $L_{\mathrm{unc}}(v) = \mathbb{E}|X_1 - X_0 - v(X_t,t)|^2$ and $t < 1$, with $q_i(x,t) = \Pr(I = i \mid X_t = x, t) \propto \pi_0\big(\frac{x - tx^i}{1-t}\big)$,*

$$v_{\mathrm{unc}}^\star(x,t) = \sum_i q_i(x,t)\,\frac{x^i - x}{1-t}, \qquad p_1^{\mathrm{unc}} = \widehat{\rho}_X = \tfrac{1}{N}\sum_i \delta_{x^i}.$$

Moreover $\inf_v L_{\mathrm{unc}} > 0$ whenever $x^i \neq x^j$ for some $i \neq j$.

Corollary 1 (The correct conditional/unconditional distinction)**.** *Both exact population flows are supported on the training set: $p_1^{\mathrm{cond}}(\cdot \mid y^i) = \delta_{x^i}$ and $p_1^{\mathrm{unc}} = \frac{1}{N}\sum_i \delta_{x^i}$. The distinction is therefore not "memorises vs. does not" but* single-example *vs.* full-empirical-measure *memorisation, with observable difference in the conditional second moment:*

$$\mathrm{Cov}[p_1^{\mathrm{cond}}(\cdot \mid y^i)] = 0 \qquad vs. \qquad \mathrm{Cov}[p_1^{\mathrm{unc}}] = \widehat{\Sigma}_X. \tag{2}$$

The often-cited observation $\operatorname{tr}\operatorname{Cov} \approx \operatorname{tr}\widehat{\Sigma}_X$ for the unconditional baseline is precisely (2), i.e. evidence of full-empirical-measure memorisation, *not* of non-memorisation.

3.2 Label smoothing is exactly kernel regression

Proposition 4 (Kernel-weighted population minimiser). *For $h > 0$, $t < 1$,*

$$v_h^\star(x,t,y) = \sum_i w_i^{(h)}(x,t,y)\,\frac{x^i - x}{1-t}, \qquad w_i^{(h)} = \frac{\pi_0\big(\frac{x-tx^i}{1-t}\big)K_h(y-y^i)}{\sum_j \pi_0\big(\frac{x-tx^j}{1-t}\big)K_h(y-y^j)}. \tag{3}$$

Proof. Given $I = i$, X_t and $\widetilde{Y}$ are conditionally independent with densities $(1-t)^{-d}\pi_0(\frac{x-tx^i}{1-t})$ and $K_h(y-y^i)$; their product is the joint conditional density. The i-independent factors $(1-t)^{-d}$, $(2\pi h^2)^{-k/2}$ and the uniform prior $1/N$ cancel in Bayes' rule, giving $\Pr(I = i \mid X_t = x, t, \widetilde{Y} = y) = w_i^{(h)}$. Given $I = i$ the target is $(x^i - x)/(1-t)$; substitute into Proposition 1. □

Equation (3) factorises the index posterior into a spatial source-consistency term $\pi_0(\frac{x-tx^i}{1-t})$ and a label kernel $K_h(y-y^i)$. With the normalised label weights $p_i^{(h)}(y) = K_h(y-y^i)/\sum_j K_h(y-y^j)$, the field (3) is the conditional velocity of the coupling $I \sim p^{(h)}(y)$, $X_0 \sim \pi_0$, $X_1 = x^I$.

Theorem 1 (Endpoint law under label smoothing). *For every $h > 0$ and query y,*

$$p_h^{\mathrm{gen}}(\cdot \mid y) = \sum_{i=1}^N p_i^{(h)}(y)\,\delta_{x^i}. \tag{4}$$

Proof. $v_h^\star$ is the conditional velocity of the mixture coupling and has the smooth form of Lemma 2, so its flow transports π_0 through the coupling's one-time marginals; its endpoint is $X_1 = x^I$ with $\Pr(I = i) = p_i^{(h)}(y)$. Take $t \uparrow 1$. □

The bandwidth interpolates continuously between the two hard regimes: if y has a unique nearest label $i^\star$ then $p^{(h)}(y) \to \mathbf{1}\{i = i^\star\}$ as $h \to 0$ (recovering Proposition 2), while $p_i^{(h)} \to 1/N$ as $h \to \infty$ (recovering p_1^{unc}).

Proposition 5 (Exact finite-N moments). *$\bar{x}_h(y) = \sum_i p_i^{(h)}(y)\,x^i$ and $\operatorname{Cov}_h(y) = \sum_i p_i^{(h)}(y)\,(x^i - \bar{x}_h(y))(x^i - \bar{x}_h(y))^\top$, both exactly computable from the training set for any h. They are the correct reference against which generated moments must be compared. $\operatorname{tr}\operatorname{Cov}_h(y)$ need not be monotone in h; monotonicity is not implied and must be reported empirically.*

Proposition 6 (Label smoothing never leaves the training set). *For every $h \in [0,\infty)$, $\operatorname{supp}(p_h^{\mathrm{gen}}(\cdot \mid y)) \subseteq \{x^1, \ldots, x^N\}$. Consequently, if the true posterior $p(\cdot \mid y)$ is absolutely continuous,*

$$W_2^2\big(p_h^{\mathrm{gen}}(\cdot \mid y),\, p(\cdot \mid y)\big) \;\geq\; \int \operatorname{dist}\big(x, \{x^i\}\big)^2\, p(x \mid y)\,dx \;>\; 0, \tag{5}$$

and this lower bound does not depend on h.

Proof. The support claim is Theorem 1. For any coupling (X, X') of p_h^{gen} and $p(\cdot \mid y)$, $X \in \{x^i\}$ a.s. so $|X - X'| \geq \operatorname{dist}(X', \{x^i\})$ pointwise; take expectations and infimise. The right side is the same for every h and is strictly positive because a finite set is $p(\cdot \mid y)$-null. □

Proposition 6 is the central caveat on the "variance is restored at $h \approx 0.1$" claim: label smoothing *redistributes weight over training atoms*; it never generates new samples. Matching $\operatorname{tr}\operatorname{Cov}_h$ matches a second moment, not the posterior. A genuine remedy requires $h \to 0$ *and* $N \to \infty$ jointly, which is the Nadaraya–Watson consistency regime (bias $O(h^2)$, variance $O((Nh^k)^{-1})$, optimal $h^\star \asymp N^{-1/(k+4)}$)—not h alone. This is a different $h \to 0$ limit from the fixed-N nearest-neighbour collapse above: same symbol, opposite conclusions.

Proposition 7 (Bandwidth expansion of the conditional covariance)**.** *Let (x^j, y^j) be i.i.d. with $\Sigma(y) = \operatorname{Cov}(X \mid Y = y)$ and $J(y) = \partial\mu/\partial y$. Under standard smoothness and $Nh^k \to \infty, h \to 0$,*

$$\operatorname{Cov}_h(y) = \Sigma(y) + h^2 J(y)J(y)^\top + O(h^4) + O_P\big((Nh^k)^{-1/2}\big), \tag{6}$$

so $\operatorname{tr}\operatorname{Cov}_h(y) = \operatorname{tr}\Sigma(y) + h^2\|J(y)\|_F^2 + \cdots$.

The between-group inflation $h^2\|J\|_F^2$ is positive and, unlike the bias of the conditional *mean*, is not annihilated by kernel symmetry; it produces the right-hand (over-smoothing) branch of the empirical U-curve. In the linear-Gaussian model $\mu(y)$ is exactly linear, so $J = \Sigma_{\text{post}}A^\top/\sigma_{\text{obs}}^2$ is known in closed form and the $O(h^4)$ term vanishes: (6) is then a parameter-free prediction.

3.3 Stochastic interpolant noise does not help

Replace the interpolant by $X_t = (1-t)X_0 + tX_1 + \gamma(t)Z$ with $Z \sim \mathcal{N}(0, I_d)$ independent and $\gamma(t) = \sigma\sqrt{t(1-t)}$, so $\gamma(0) = \gamma(1) = 0$. A single Z defines the whole trajectory, and the correct pathwise-derivative target is

$$\dot{X}_t = X_1 - X_0 + \dot{\gamma}(t)Z, \qquad \dot{\gamma}(t) = \frac{\sigma(1-2t)}{2\sqrt{t(1-t)}}. \tag{7}$$

Regressing on $X_1 - X_0$ alone is a *different* objective and does not yield the marginal velocity of the noisy path. Let $s_t^2 := (1-t)^2 + \gamma(t)^2 = (1-t)[(1-t) + \sigma^2 t]$, so $s_0 = 1$, $s_1 = 0$.

Proposition 8 (Closed form and endpoint invariance in σ)**.** *Let $h = 0$ with distinct labels and $v_\sigma^\star = \mathbb{E}[\dot{X}_t \mid X_t, t, Y]$ for target (7). Then for $t < 1$: (a) $v_\sigma^\star(x, t, y^i) = x^i + c(t)(x - tx^i)$ with $c(t) = \frac{-(1-t)+\frac{\sigma^2}{2}(1-2t)}{s_t^2} = \frac{1}{2}\frac{d}{dt}\log s_t^2$; (b) the flow is $x_t = t\,x^i + s_t\,x_0$; and (c) since $s_1 = 0$,*

$$p_1^{\text{cond}}(\cdot \mid y^i) = \delta_{x^i} \quad \textit{for every } \sigma \geq 0. \tag{8}$$

Proof sketch. Given $I = i$, $r := x - tx^i = (1-t)X_0 + \gamma(t)Z \sim \mathcal{N}(0, s_t^2 I_d)$; the Gaussian conditional means $\mathbb{E}[X_0 \mid r] = \frac{1-t}{s_t^2}r$, $\mathbb{E}[Z \mid r] = \frac{\gamma}{s_t^2}r$ give (a) after $\dot{\gamma}\gamma = \frac{\sigma^2}{2}(1-2t)$. For (b), $u_t := x_t - tx^i$ solves $\dot{u}_t = c(t)u_t$, so $u_t = s_t x_0$. (c) $s_1 = 0$. □

Corollary 2 (What interpolant noise does and does not change)**.** *Interpolant noise does not change the population endpoint law (8) for any σ; it* does *raise the irreducible error ($\inf_v L_\sigma > 0$ for $\sigma > 0$,* whereas *$\inf_v L_0 = 0$) and halves the Lipschitz blow-up rate: $\lim_{t\uparrow 1}(1-t)|c(t)| = 1$ for $\sigma = 0$ and $\frac{1}{2}$ for $\sigma > 0$.*

Proposition 9 (The two mechanisms act on different factors)**.** *In the factorisation (3), label smoothing modifies the label factor $K_h(y - y^i)$ and therefore changes p_1 (Theorem 1); interpolant noise rescales the spatial factor $\mathcal{N}(tx^i, s_t^2 I_d)$* isotropically and identically for every i*, leaving the endpoint law unchanged (Proposition 8). The two remedies are therefore not interchangeable.*

This cleanly separates the present conditional mechanism from the unconditional "resample-x_0" remedy of the gradient-variance line [gra, 2025], which operates through attainability rather than through the population optimum.

3.4 Finite capacity: a genuine representation floor

Proposition 10 (Lipschitz lower bound)**.** *Fix i, $t < 1$, $f_i(x,t) = (x^i - x)/(1-t)$, and suppose $v(\cdot, t, y^i)$ is L-Lipschitz in x. Under $X_t \mid I = i \sim \mathcal{N}(tx^i, (1-t)^2 I_d)$, $\mathbb{E}[|v(X_t, t, y^i) - f_i(X_t, t)|^2 \mid I = i] \geq d\,[1 - L(1-t)]_+^2$.*

Corollary 3 (Loss floor under a uniform Lipschitz constraint)**.** *For $\mathcal{F}_L = \{v : \mathrm{Lip}_x\, v(\cdot, t, y^i) \leq L\ \forall t, i\}$ with $L \geq 1$ and $t \sim \mathrm{Unif}(0,1)$,*

$$\inf_{v \in \mathcal{F}_L} L_0(v) \ \geq\ d \int_{1-1/L}^{1} [1 - L(1-t)]^2\, dt \ =\ \frac{d}{3L} \ >\ 0. \tag{9}$$

Two warnings temper (9). First, $L^\star_{\text{model}} > 0$ requires the constraint: an MLP with unbounded weights lies in no fixed $\mathcal{F}_L$, and by universal approximation $\inf_\theta L(v_\theta)$ may be 0. The correct statement is conditional: any architecture with $\mathrm{Lip}_x\, v_\theta \leq L$ cannot beat $d/(3L)$. Second, in our experiments ($d = 2$, plateau $L_{\text{trained}} \approx 0.36$) the bound demands only $L \geq 1.85$, whereas a trained network has Lip_x of order 10^1–10^2, giving a floor of order 10^{-2}—one to two orders of magnitude below the plateau. Corollary 3 is thus best used as a *measurement*: estimate L, compare $d/(3L)$ to L_{trained}, and report which gap dominates (Section 4.4).

3.5 Scope: what is and is not predicted

The population theory proves exact consequences for the empirical minimiser (Propositions 2–9, Corollary 3). It does *not* determine: the number of SGD steps to approach the optimum; the dependence of *observed* collapse on N at fixed architecture (the population optimum collapses for every finite N, so a non-collapsing N=5000 run probes representation/optimisation, it does not contradict Proposition 2); the dependence of observed collapse on σ_{obs} (the theory implies no monotonicity, so a flat sweep is *consistent with*, not a falsification of, the theory); the monotonicity of $\mathrm{tr}\,\mathrm{Cov}_h$ in h; or finite-sample error bars.

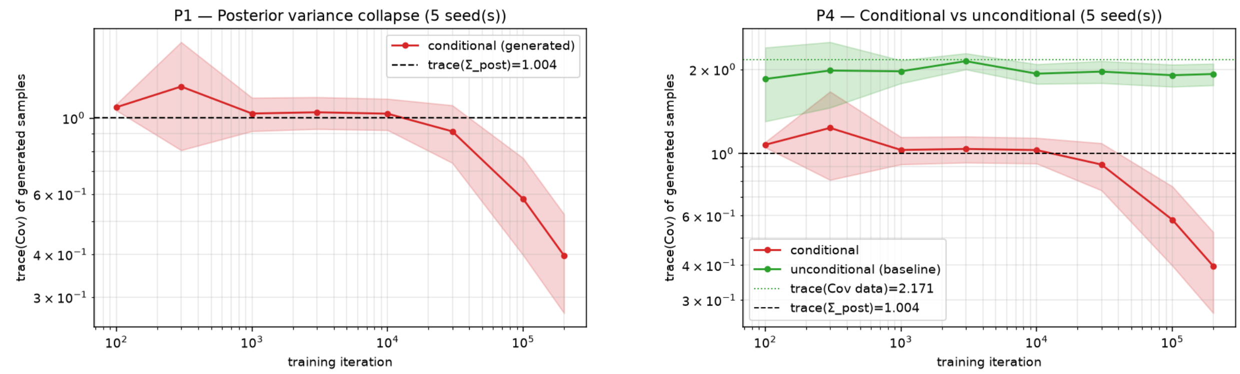
> *Phrasing rule. A quantity in the second list is never a "theoretical prediction" that "matched" or "failed." Correct form: the exact population theory does not determine X; the observed behaviour of X is an empirical finding. We follow this rule in Section 4, marking such results "(empirical)."*

4 Experiments

Setup. Unless noted, EXP-1 uses the linear-Gaussian problem d=2, k=1, N=200, σ_{obs}=0.1, prior and source $\mathcal{N}(0, I)$, deterministic linear interpolant (σ=0); a 4-layer width-128 MLP with SiLU and a sinusoidal time embedding (dim 64); Adam at lr=10^{-3}, batch 256; RK4 integration with 100 steps stopping at t=1−10^{-3}; evaluation over M=1000 samples at 20 training conditions and 8 held-out conditions. We do *not* early-stop: overtraining is the object of study. Sanity checks pass first (analytic posterior satisfies the normal equations to 1.8×10^{-15}; closed-form field integration maps every $x_0 \to x^i$; the unconditional model does not leak y), so the collapse below is not an artefact of a y-leak or a broken ODE. Parameter sweeps use five seeds per point (seeds 0–2 on CPU, seeds 3–4 on 2×T4 GPU via the same training code, `device: auto`).

4.1 EXP-1: exact-theory quantities (P1–P4)

Figure 1 summarises the four exact-theory observables. As training proceeds past the good-calibration phase, the conditional generated variance collapses from $\approx \Sigma_{\text{post}}$ toward 0 (P1), the



(a) P1: conditional variance collapse. (b) P4: conditional vs. unconditional.

Figure 1: Overtraining collapses the conditional generated variance toward zero, while the unconditional model retains data-scale variance (Corollary 1).

	iteration				
quantity (t=200k unless noted)	10^2	10^4	10^5	2×10^5	7×10^5
conditional $\operatorname{tr}\operatorname{Cov}$	1.10	1.06	0.62	0.396 ± 0.127	0.16
velocity rel. error vs. ($\star$)	0.72	—	0.53	0.40 ± 0.05	0.29
$\lVert\text{mean} - x^i\rVert$	1.02	—	0.59	0.38 ± 0.10	0.20
$\lVert\text{mean} - \mu_{\text{post}}\rVert$	0.07	—	—	0.66	—
unconditional $\operatorname{tr}\operatorname{Cov}$	1.87	1.95	1.95	1.926 ± 0.171	—

Table 1: EXP-1 (mean±std over 5 seeds where shown; extended run is seed 0). Reference values: $\operatorname{tr}\Sigma_{\text{post}} \approx 1.004$, $\operatorname{tr}\widehat{\Sigma}_X = 2.171$. All four conditional quantities descend together toward the δ_{x^i} limit; the extended seed-0 run reaches $\operatorname{tr}\operatorname{Cov} = 0.16$ at 7×10^5 and is still decreasing.

velocity field approaches the collapsed closed form $(x^i - x)/(1-t)$ (P2), the generated mean moves from μ_{post} toward the training point x^i (P3), and the unconditional baseline retains data-scale variance throughout (P4).

The variance tracks the true posterior up to $\sim 10^4$ iterations—the model is a correct Bayesian sampler in this phase, which is exactly why early stopping works [phy, 2026]—and then collapses. At 2×10^5 the variance has not yet reached 0 but stalls near 0.40; the extended seed-0 run (Fig. 2) shows all four observables decreasing together with the loss ($0.71 \to 0.36$), so the residual is an *optimisation* gap, not a failure of Proposition 2. At a fixed learning rate the run eventually diverges near 10^6 iterations (the target ($\star$) steepens as $t \to 1$); the best collapsed checkpoint is $\approx 7 \times 10^5$.

4.2 Scope-marked sweeps (P5, P6, d/k; empirical)

The following depend on N and σ_{obs} at fixed architecture and are *not* determined by the population theory (Section 3.5); we report them as empirical findings using the collapse ratio $\operatorname{tr}\operatorname{Cov} / \operatorname{tr}\Sigma_{\text{post}}$ at 2×10^5 iterations.

4.3 Label smoothing tracks the kernel reference (P7)

This is the strongest confirmation in the project. The correct reference is Cov_h (Theorem 1, Proposition 5), *not* Σ_{post}. Table 4 evaluates the trained checkpoints against the exact kernel

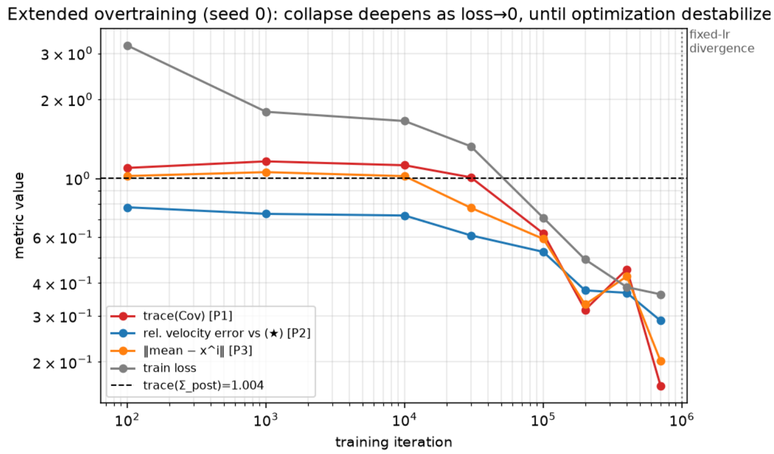


Figure 2: Extended seed-0 trajectory: $\operatorname{tr}\mathrm{Cov}$, velocity error, $\|\mathrm{mean} - x^i\|$ and the training loss decrease together, identifying P1–P3 as one phenomenon driven by loss $\to 0$ (the exact minimiser has loss 0).

N	50	200	1000	5000
ratio	0.049	0.389	0.912	0.990
$\pm$	0.038	0.123	0.103	0.031

(a) P6: collapse vs. dataset size.

(d,k)	$(2,1)$	$(10,1)$	$(10,10)$
ratio	0.390	0.317	0.0065
$\pm$	0.127	0.033	0.0024

(b) Collapse vs. dimension/observation.

Table 2: 5-seed sweeps (empirical). P6 is monotone and stable across seeds: small N collapses almost completely, N=5000 barely collapses ($\approx$ true posterior)—a representation/optimisation statement, not a contradiction of Proposition 2. Larger k (a more informative label) sharpens the identifier and deepens collapse, consistent with the identification mechanism.

moments over 20 conditions (M=1000), across five seeds.

We correct an earlier "perfect restoration at $h \approx 0.1$" reading: $\operatorname{tr}\mathrm{Cov}$ merely crosses $\Sigma_{\text{post}} \approx 1.02$ near h=0.1 because for this problem $\Sigma_{\text{post}} \approx \mathrm{Cov}_h\,|_{h=0.1} \approx 1.00$—a coincidence. The correct target is Cov_h, which the model attains at *every* h, not only h=0.1. Figure 3 and Table 5 confirm the field itself: the learned v_θ matches the kernel minimiser (3) far better than the single-atom field ($\star$), and the gap widens with h, exactly as the mixture structure predicts.

Atomicity caveat (empirical confirmation of Proposition 6). Even at the h that best matches Σ_{post}, the generated law remains atomic. The effective number of atoms is only $n_{\text{eff}} \approx$ 26/200 at h=0.1 and $\approx$ 102/200 at h=0.5 (Table 4). Directly measuring the discrepancy to the analytic posterior (Section 4.4) shows the MMD decreasing with h but never reaching zero at any h. Matching variance is not recovering the posterior.

Interpolant noise does not help (Proposition 8). Table 6 adds stochastic interpolant noise. With the *corrected* target (7) (verified by four closed-form checks), the generated variance is unchanged from the buggy $X_1 - X_0$ target and far below the posterior— exactly as (8) predicts: the endpoint law is δ_{x^i} for every σ, independently even of whether the target is correct. Only label noise, acting on the *label* factor, can change p_1 (Proposition 9).

σ_{obs}	0.01	0.1	0.5	1.0
$\operatorname{tr}\Sigma_{\text{post}}$	1.000	1.018	1.266	1.531
generated $\operatorname{tr}\operatorname{Cov}$	0.472 ± 0.108	0.396 ± 0.127	0.416 ± 0.155	0.507 ± 0.236
collapse ratio	0.472	0.390	0.328	0.331

Table 3: P5 (empirical), 5 seeds. The between-σ_{obs} spread in the ratio (0.33–0.47, range ~ 0.14) is *smaller* than the between-seed std at each point (0.11–0.24): the sweep is flat within seed noise. This is consistent with an identification-through-y mechanism that operates for every $\sigma_{\text{obs}} > 0$; the theory implies no monotonicity, so flatness is not a falsification.

h	Cov_h (Thm 1)	Σ_{post}	generated $\operatorname{tr}\operatorname{Cov}$	**ratio-to-kernel**	ratio-to-post	$n_{\text{eff}}/200$
0.01	0.583	1.018	0.673 ± 0.187	$1.308 \pm 0.376^{\dagger}$	0.66	3.5
0.05	0.921	1.018	0.926 ± 0.169	$\mathbf{1.002 \pm 0.050}$	0.91	13.6
0.10	1.004	1.018	1.001 ± 0.173	$\mathbf{0.994 \pm 0.039}$	0.98	26.0
0.50	1.281	1.018	1.294 ± 0.166	$\mathbf{1.011 \pm 0.022}$	1.27	101.5

Table 4: P7 kernel reference, 5 seeds. For every $h \geq 0.05$ the generated variance tracks the exact kernel target Cov_h to within seed noise (ratio-to-kernel ≈ 1.00, std ≤ 0.05)—a verification of the *endpoint law*, stronger than a velocity-field check. †At h=0.01 ($n_{\text{eff}} \approx 3.5$, single-atom regime) Cov_h is near-degenerate for many conditions, so the per-condition ratio is unstable and high; the ratio of mean traces 0.673/0.583=1.15 is closer to 1.

4.4 Additional verification (gap to optimum, Lipschitz, posterior distance)

Three checkpoint-only measurements corroborate the theory. (i) *Optimality gap.* The Monte-Carlo gap $L(v_\theta) - L(v_h^\star)$, with $L(v_h^\star) = \mathbb{E}[\operatorname{Var}(U \mid X_t, t, \widetilde{Y})]$ computed from the exact minimiser, is non-negative at every checkpoint and decreases monotonically toward 0 at every h (e.g. h=0.1: 1.14 $\to$ 0.22 $\to$ 0.102 at $10^2/3{\times}10^4/2{\times}10^5$ steps), consistent with ratio-to-kernel $\to 1$. The irreducible error $L(v_h^\star)$ increases with h ($0 \to 1.88$): the price of smoothing. (ii) *Lipschitz floor.* The empirical $\operatorname{Lip}_x v_\theta$ grows from 7.1 at t=0.5 to 65.7 at t=0.99; the Corollary 3 floor $d/(3L) \approx 0.010$ is $\sim 50\times$ below the observed plateau 0.48, so the plateau is optimisation-limited, exactly as anticipated. (iii) *Posterior distance.* The MMD to the analytic posterior $\mathcal{N}(\mu_{\text{post}}, \Sigma_{\text{post}})$ falls with h ($0.256 \to 0.012$) but never reaches 0, and the Sinkhorn distance plateaus near 2.7—a direct numerical witness of the h-independent floor (5).

4.5 EXP-2 and EXP-3: selective memorisation beyond the Gaussian case

EXP-2 (Gaussian mixture). A 4-mode GMM prior with a rank-deflating linear observation gives a genuinely *bimodal* analytic posterior. As the loss descends, mode coverage falls 1.00 $\to$ 0.72 (toward 0.5, i.e. one of the two modes) and the MMD to the true posterior grows $\sim 40\times$; samples that initially cover both modes collapse onto x^i in a single mode and abandon the other—selective memorisation (Fig. 4, left).

EXP-3 (MNIST inpainting). Masking the lower half of 32×32 MNIST (N=500) with a small U-Net, we draw 16 completions per condition. The inpainted-region pixel variance drops $\sim 100\times$ ($0.140 \to 0.0013$) while the distance to the nearest training image falls to 0.0005 and the observed-region reconstruction stays exact (2×10^{-5}). Qualitatively, diverse completions at iteration 200

h	rel. error vs. **kernel** (3)	rel. error vs. ($\star$)	TV of atom mixture
0.01	$\mathbf{0.318 \pm 0.030}$	0.599 ± 0.104	0.31 ± 0.18
0.05	$\mathbf{0.195 \pm 0.019}$	0.683 ± 0.103	0.22 ± 0.02
0.10	$\mathbf{0.162 \pm 0.021}$	0.703 ± 0.092	0.16 ± 0.04
0.50	$\mathbf{0.164 \pm 0.025}$	0.791 ± 0.070	0.16 ± 0.02

Table 5: Velocity-field match (5 seeds). v_θ is 2–4× closer to the kernel field than to the collapsed field at every h. Assigning generated samples to the nearest atom recovers the weights $p_i^{(h)} \propto K_h$ with total variation 0.16–0.31. At h=0.5 the generated variance 1.29 *overshoots* Σ_{post}=1.02, matching the $+h^2\|J\|_F^2$ term of Proposition 7 ($\|J\|_F^2 = 0.4031$, exact).

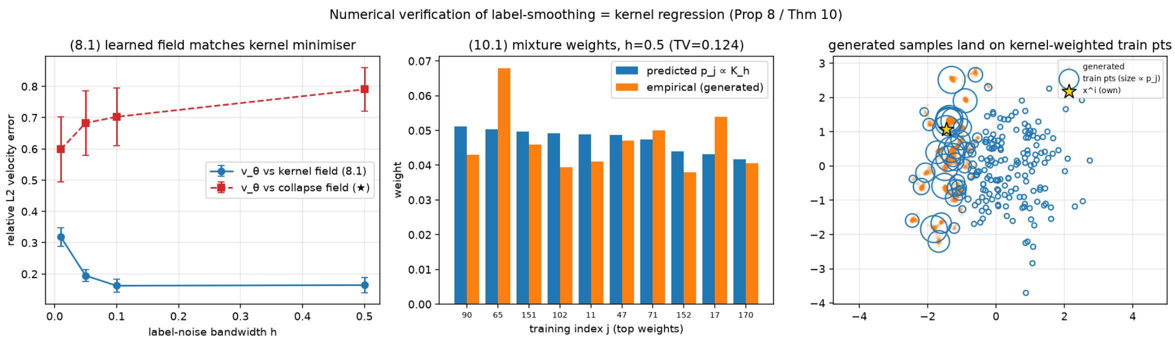


Figure 3: Kernel-theory verification: the learned field matches the kernel minimiser (3) (left); the predicted atom weights $p_i^{(h)} \propto K_h$ match the empirical nearest-atom assignment (centre); and the generated trace-covariance tracks $\operatorname{tr}\operatorname{Cov}_h$ across h (right).

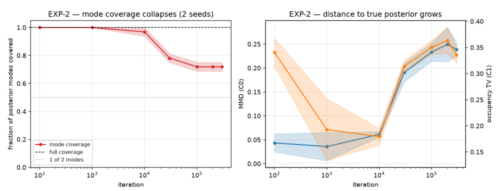
become identical by iteration 15000 and coincide with the nearest training image (Fig. 4, right)—the same collapse on real image data, reached earlier because the loss converges deeper here.

5 Limitations

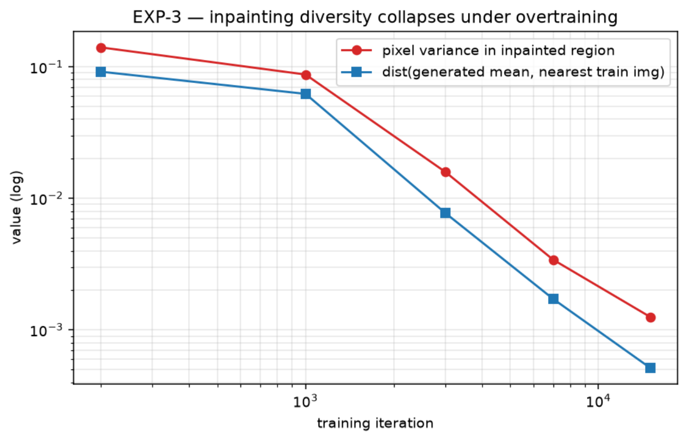
The theory characterises the exact population minimiser; a finite network trained by SGD approaches it only up to an optimisation gap, which is why hard-conditioning collapse is *optimisation-paced* and only partial within a finite budget (for $h > 0$ the model does reach the population optimum Cov_h). At a fixed learning rate, extreme overtraining eventually diverges near 10^6 iterations; reaching total collapse cleanly would need a learning-rate schedule. The sweeps P5 and P6 are empirical findings the population theory does not determine and must not be read as prediction matches. The atomicity obstruction (Proposition 6) means the label-smoothing "remedy" matches moments but never recovers a continuous posterior. Finally, EXP-3 uses a single seed at N=500; broader image datasets, mask types, and an N sweep are left to future work. Checkpoints are regenerable and not tracked in version control, so reproducing the checkpoint-based tables from a clean clone requires retraining the *p7y* runs (deterministic per seed); all sweep tables reproduce directly from the tracked metrics.

interpolant σ	0.1	0.3
$\operatorname{tr}\operatorname{Cov}$, old target $X_1 - X_0$ (2 seeds)	0.372 ± 0.140	0.429 ± 0.121
$\operatorname{tr}\operatorname{Cov}$, corrected target (7) (5 seeds)	$\mathbf{0.367 \pm 0.146}$	$\mathbf{0.383 \pm 0.131}$
mean bias	0.68	0.66

Table 6: Interpolant-noise contrast. Correcting the target leaves the result unchanged within seed noise—a confirmation of Proposition 8(c), not a failure.



(a) EXP-2: mode coverage → single mode.



(b) EXP-3: inpainting variance / NN-distance collapse.

Figure 4: Selective memorisation beyond the linear-Gaussian model: a bimodal GMM posterior (left) and MNIST inpainting (right) both collapse onto memorised training content as the loss descends.

6 Conclusion

Conditional collapse in flow matching is the zero-bandwidth limit of a kernel-weighted empirical conditional flow. Under hard conditioning with identifying labels the exact population minimiser has zero conditional variance; label smoothing turns this into exact kernel regression over the training atoms, with a parameter-free moment reference that a trained network tracks to within seed noise across five seeds. The two apparent remedies are provably different—label noise moves the population endpoint, interpolant noise cannot—and even the working remedy only matches moments, bounded away from the true posterior by an h-independent Wasserstein floor. The same collapse holds on a bimodal GMM posterior and on MNIST inpainting.

Reproducibility. All numbers are produced by the released code. Theory checks: `test_kernel_theory.py` (reference table to $< 0.05\%$), `analyze_p7_kernel.py`, `verify_kernel_theory.py`, `verify_prop17.py`. Experiments: `run_exp1.sh`, `run_sweeps.py` (+`--gpus N`), `train_exp2`, `train_exp3`, each followed by the matching analysis script.

References

Gradient variance reveals failure modes in flow-based generative models. *arXiv preprint arXiv:2510.18118*, 2025. Unconditional memorization via the deterministic interpolant; the resample-x_0 mechanism.

Solving physics-constrained inverse problems with conditional flow matching. *arXiv preprint arXiv:2603.14135*, 2026. Source of the observed variance-collapse / selective-memorization phenomenon.

Giulio Biroli and Marc Mézard. Generative diffusion in very large dimensions. *Journal of Statistical Mechanics: Theory and Experiment*, 2023(9):093402, 2023. arXiv:2306.03518.

Xiangming Gu, Chao Du, Tianyu Pang, Chongxuan Li, Min Lin, and Ye Wang. On memorization in diffusion models. *arXiv preprint arXiv:2310.02664*, 2023.

Zahra Kadkhodaie, Florentin Guth, Eero P. Simoncelli, and Stéphane Mallat. Generalization in diffusion models arises from geometry-adaptive harmonic representations. In *International Conference on Learning Representations (ICLR)*, 2024. Oral.

Mason Kamb and Surya Ganguli. An analytic theory of creativity in convolutional diffusion models. *arXiv preprint arXiv:2412.20292*, 2024. ICML 2025.

Jakiw Pidstrigach. Score-based generative models detect manifolds. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2206.01018.

A Deferred proofs

Lemma 1 (Change of variables)**.** *Fix i and $t < 1$. The map $X_0 \mapsto X_t = (1-t)X_0 + tx^i$ is invertible affine with $X_0 = (X_t - tx^i)/(1-t)$, so $\sigma(X_t \mid I = i) = \sigma(X_0 \mid I = i)$ and the conditional density is $p(x \mid I = i, t) = (1-t)^{-d}\pi_0\big(\frac{x-tx^i}{1-t}\big)$. The Jacobian $(1-t)^{-d}$ is i-independent and cancels in every posterior over I. Because π_0 is Gaussian, $p(x \mid I = i, t) > 0$ for every x and $t < 1$.*

Lemma 2 (Well-posedness)**.** *Let $w_1, \dots, w_N : \mathbb{R}^d \times [0,1) \to [0,1]$ be C^∞ with $\sum_i w_i \equiv 1$ and $v(x,t) = \sum_i w_i(x,t)(x^i - x)/(1-t)$. Then v is C^∞ hence locally Lipschitz on $\mathbb{R}^d \times [0,1)$, the ODE $\dot{x}_t = v(x_t, t)$ has a unique solution on $[0,1)$ for every initial condition, and with $M = \max_i |x^i|$, $|x_t| + M \leq (|x_0| + M)/(1-t)$. Consequently the continuity equation has a unique measure solution and endpoint laws are the weak limits $p_1 = \lim_{t\uparrow 1} p_t$.*

Both $v^\star_{\text{unc}}$ and $v^\star_h$ have this form with smooth weights, so Lemma 2 applies and no "assume the flow is well defined" hypothesis is needed. Proofs of Propositions 3, 7, 10 and Corollary 3 follow the arguments in the theory notes accompanying the code and are omitted here for brevity.